

ADVANCED EARTH SCIENCES & ENVIRONMENTAL GEODYNAMICS

New Egyptian Baccalaureate Framework (EG-Bac Track) - Grade 12

ACADEMIC YEAR: 2025 / 2026 | EG-BAC GRADE 12 (ADVANCED NATURAL SCIENCES & EARTH SYSTEMS)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced collegiate-level earth sciences focusing on quantitative seismic wave velocities, paleomagnetic polar reversals, crystallographic unit cells, and thermodynamic geochemical balances.

CORE CURRICULUM HIGHLIGHTS:

- Mathematical seismic wave arrival time modeling for epicenter localization and Mohorovičić discontinuity mapping
- X-ray crystallography Bragg law applications and Miller indices for mineral lattices
- Geodynamic mantle convection simulation and Wilson cycle basin evolution
- Global carbon cycle quantification, marine upwelling dynamics, and planetary boundaries

APPROVED CHAPTER MODULES:

- Ch 1: Lithospheric Structure, Seismic Tomography & Geologic Formations	pp. 1-40 (15 core units)
- Ch 2: Mineral Crystallography, X-ray Diffraction & Geochemistry	pp. 41-72 (12 core units)
- Ch 3: Magmatic Petrology, Isostasy & Metamorphic Facies	pp. 73-114 (18 core units)
- Ch 4: Wilson Cycle, Plate Kinematics & Paleomagnetism	pp. 115-154 (16 core units)
- Ch 5: Quantitative Geomorphology, Hydrologic Modeling & Weathering	pp. 155-190 (16 core units)
- Ch 6: Marine Geophysical Dynamics & Extreme Ecosystem Energetics	pp. 191-220 (14 core units)
- Ch 7: Earth Systems Modeling, Biogeochemical Cycles & Sustainability	pp. 221-248 (14 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: EG-Bac Earth Systems, Crystallography & Geophysics Compendium (40p - 4.1 MB)
Full Ministerial Textbook: 248 Pages - 34.2 MB (Direct on MOE Portal)
Publisher: Egyptian Ministry of Education - National Baccalaureate Directorate
Official Portal: <https://moe.gov.eg/ar/elearning/egbac/geology/> - Code: EGBAC-GEOL-G12
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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced collegiate-level earth sciences focusing on quantitative seismic wave velocities, paleomagnetic polar reversals, crystallographic unit cells, and thermodynamic geochemical balances. Mathematical seismic wave arrival time modeling for epicenter localization and Mohorovičić discontinuity mapping X-ray crystallography Bragg law applications and Miller indices for mineral lattices Geodynamic mantle convection simulation and Wilson cycle basin evolution Global carbon cycle quantification, marine upwelling dynamics, and planetary boundaries

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Ch 1: Lithospheric Structure, Seismic Tomography & Geologic Formations	4 wks	9 pts	14.3%	15 core instructional units covering national standards and
Ch 2: Mineral Crystallography, X-ray Diffraction & Geochemistry	4 wks	9 pts	14.3%	12 core instructional units covering national standards and
Ch 3: Magmatic Petrology, Isostasy & Metamorphic Facies	4 wks	9 pts	14.3%	18 core instructional units covering national standards and
Ch 4: Wilson Cycle, Plate Kinematics & Paleomagnetism	4 wks	9 pts	14.3%	16 core instructional units covering national standards and
Ch 5: Quantitative Geomorphology, Hydrologic Modeling & Weathering	4 wks	9 pts	14.3%	16 core instructional units covering national standards and
Ch 6: Marine Geophysical Dynamics & Extreme Ecosystem Energetics	4 wks	9 pts	14.3%	14 core instructional units covering national standards and
Ch 7: Earth Systems Modeling, Biogeochemical Cycles & Sustainability	4 wks	6 pts	14.3%	14 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CH 1: LITHOSPHERIC STRUCTURE, SEISMIC TOMOGRAPHY & GEOLGIC FORMATIONS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 1: Lithospheric Structure, Seismic Tomography & Geologic Formations.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Ch 1: Lithospheric Structure, Seismic Tomography & Geologic Formations.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 2: MINERAL CRYSTALLOGRAPHY, X-RAY DIFFRACTION & G

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 2: Mineral Crystallography, X-ray Diffraction & Geochemistry.
- [+] Synthesize and apply analytical frameworks across all 12 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Ch 2: Mineral Crystallography, X-ray Diffraction & Geochemistry.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 3: MAGMATIC PETROLOGY, ISOSTASY & METAMORPHIC FACIES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 3: Magmatic Petrology, Isostasy & Metamorphic Facies.
- [+] Synthesize and apply analytical frameworks across all 18 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Ch 3: Magmatic Petrology, Isostasy & Metamorphic Facies.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

ADVANCED EARTH SCIENCES & ENVIRONMENTAL GEODYNAMICS	PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$	Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Equilibrium Balance Criterion: $\sum F_{net} = 0, \sum M_{pivot} = 0$	Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$	Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Normalized Standard Score: $Z = (X - \mu) / \sigma$	Curricular Efficiency Ratio: $\eta = (\text{Output}_{actual} / \text{Input}_{standard}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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